

Paper Corpus Generation: An End-to-End Account

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Abstract

This note follows one case from a random seed to the arrays read by the trainer. The replacement corpus contains four kinds of periodic water-wave states: Stokes waves, Tanaka-profile sums, Benjamin–Feir perturbations, and JONSWAP/TMA random seas. Every label is computed with the same fixed discrete Dirichlet–Neumann operator.

A draw must first lie in the stated parameter range of its family. A time-dependent draw is then retained only if one fixed-step calculation both returns a complete finite trajectory and keeps the free surface above the bottom. No rule based on visual roughness, sign changes, slope, Hamiltonian drift, or model error removes a case. This document is the specification of the replacement corpus; the corpus itself has not yet been generated.

1 Objects and guiding rules

The horizontal interval is $0 \leq x < L$, with periodic endpoints and $L = 2\pi$, and gravity is set to one. A *case* is one sampled initial surface elevation $\eta_0(x)$, one sampled value $\xi_0(x)$ of the velocity potential at the surface, and one fixed water depth $h > 0$. A *trajectory* is the computed evolution of that case. A *row* is one retained time from an accepted case.

The case, rather than the row, is the unit of splitting, rejection, and training weight. This matters because an accepted Stokes case stores one row, whereas accepted Tanaka and Benjamin–Feir cases each store 200 rows. Counting all stored rows equally would therefore give a long trajectory much more influence than a static wave.

The construction follows four elementary rules.

1. Fix the numerical meaning of a label before sampling any cases.
2. Assign a case to training, validation, or test before seeing whether its numerical calculation succeeds.
3. Replace a failed attempt only within its original parameter category.
4. Select stored times only after the entire case has been accepted.

2 The common numerical label

For given η , ξ , and h , let $\varphi(x, y)$ solve

$$\varphi_{xx} + \varphi_{yy} = 0, \quad \varphi_y(x, -h) = 0, \quad \varphi(x, \eta(x)) = \xi(x). \quad (1)$$

Subscripts denote partial derivatives. The physical Dirichlet–Neumann operator is

$$G(\eta; h)\xi = (\varphi_y - \eta_x \varphi_x)_{y=\eta(x)}. \quad (2)$$

It is the slope-corrected upward derivative of the solution φ at the free surface. In the Zakharov equations it is also η_t , the instantaneous rate of change of the surface.

The dataset does not claim to evaluate (2) exactly. Instead, it freezes one discrete reference calculation. All fields use $N = 1024$ equally spaced points $x_\ell = 2\pi\ell/1024$, $\ell = 0, \dots, 1023$. Since the period is 2π , the Fourier wavenumbers are the integers. Let P_{128} discard all modes with $|k| > 128$, and let P_{128}° additionally remove the spatial mean. Let $\mathcal{G}_6^{1024,8}$ denote the implemented order-six Craig–Sulem calculation on 1024 points with padding factor eight. The label is

$$\boxed{G_{\text{ref}}(\eta; h)\xi = P_{128}^\circ \left[\mathcal{G}_6^{1024,8}(P_{128}\eta; h)(P_{128}\xi) \right]}. \quad (3)$$

The calculation is performed in 64-bit arithmetic. Only the final spatial arrays are converted to 32-bit form for storage. Thus every retained row is

$$(P_{128}\eta(t), P_{128}\xi(t), h, t, G_{\text{ref}}(\eta(t); h)\xi(t)). \quad (4)$$

The model is trained to reproduce the fixed discrete quantity in (3). Approximation of the continuum operator (2) is a separate question requiring convergence tests in truncation order, grid size, and padding. Every constructor removes the spatial mean of ξ ; this fixes the irrelevant freedom to add a constant to the potential.

3 One case from beginning to end

The complete production path is:

1. Assign a family, parameter category, data split, and deterministic random-number key, then draw and record all initial-condition parameters.
2. Check the family’s stated parameter rules and construct (η_0, ξ_0, h) . Before rollout, the assembled initial arrays must be finite, $h > 0$, and $h + \eta_0 > 0$ on the grid.
3. For a Stokes case, evaluate (3) at $t = 0$. For the other families, run the single fixed-step calculation in Section 5.
4. Apply the whole-case decision in Section 5. A failed trajectory contributes no rows, including no apparently good prefix.
5. After acceptance, choose the retained times and write their arrays, numerical diagnostics, and checksum.
6. Count an accepted case toward its original parameter category. If it failed, draw a new case in that same category.
7. When every family and split quota is complete, build a checked index into the array files and pass that index to the trainer.

The sampled specification is written before numerical work, so an interrupted case can be replayed without drawing a replacement. An unexpected construction error, including failure of the initial-array conditions above, stops the run; it is not silently relabeled as an ordinary rejected draw. Checksums detect missing or changed array files.

4 How the four initial-condition families are sampled

A *parameter category* is a pre-set part of one family’s parameter range that receives its own accepted-case count. The categories keep important regimes from disappearing by chance. Uniform sampling on $[a, b]$ means that every interval of the same length is equally likely. Log-uniform sampling means that $\log h$ is uniform. A uniform simplex split of a positive number A chooses nonnegative weights w_j with $\sum_j w_j = 1$ and sets the pieces to Aw_j .

Table 1: The four training families. A horizon is needed only for a time-dependent family.

Family	Categories	Main distinction	Horizon / retained rows
Stokes	4	finite/deep branch and low/moderate steepness	static / 1
Tanaka	11	regime, number of crests, and number moving right	$T = 200$ / 200
Benjamin–Feir	66	one feasible carrier–sideband integer pair per category	$T = 200$ / 200
JONSWAP/TMA	27	shallow/finite/deep, peak enhancement, and direction balance	last 0.08-grid time not exceeding 16 peak periods / 16

4.1 Fifth-order Stokes states

A Stokes draw first fixes one of four parameter categories:

$$\{\text{finite, deep}\} \times \{0.005 \leq ka < 0.03, \quad 0.03 \leq ka \leq 0.15\}.$$

Here k is the carrier wavenumber and a is the expansion-amplitude parameter of the fifth-order formula. Because the first harmonic receives nonlinear corrections, a is not exactly the realized first-harmonic amplitude. The common parameter interval is

$$0.000766 \leq a \leq 0.011494.$$

For the finite-depth branch, the mode number is $n \in \{14, \dots, 26\}$, the wavenumber is $k = n$, and

$$0.02 \leq h \leq 1.5, \quad 0.5 \leq kh \leq 5.$$

For the deep branch,

$$n \in \{1, \dots, 20\}, \quad 4 \leq h \leq 50, \quad kh \geq 5.$$

The program lists the modes compatible with the chosen category and selects one uniformly. It then draws h log-uniformly in the compatible depth interval, draws a phase ϕ uniformly on $[0, 2\pi)$, and draws a uniformly from the intersection of the common amplitude interval and the chosen ka interval. The code evaluates Fenton's fifth-order analytic Stokes expansion, using its finite-depth coefficients or deep-water limit, to construct (η_0, ξ_0) [1]. The parameter ranges, four parameter categories, random phase, and depth law above are corpus choices rather than claims made by Fenton.

The finite-depth formula is used only in its intended Stokes regime. For phase $\theta = kx + \phi$, write its elevation as

$$\eta_5(\theta) = \sum_{r=1}^5 E_r \cos(r\theta),$$

where E_r is the coefficient of harmonic r . Let H_{4096} be the largest sampled value of η_5 minus its smallest sampled value on 4096 equally spaced phases, and define

$$H_+ = H_{4096} + \frac{1}{4} \left(\frac{2\pi}{4096} \right)^2 \sum_{r=1}^5 r^2 |E_r|.$$

The last term bounds the possible height missed between two phase samples. With wavelength $\lambda = 2\pi/k$, the required support condition is

$$\text{Ur}_+ = \frac{H_+ \lambda^2}{h^3} \leq 26. \quad (5)$$

This is a conservative Ursell-number restriction for the fifth-order formula, not a fitted predictor of whether a numerical rollout would survive [1, 2]. If (5) fails, the mode, depth, phase, and parameter category remain fixed and only the amplitude is redrawn, with at most 1000 redraws. Exhausting that limit gives a zero-row attempt, and the category receives a new case. A valid Stokes case is static: it stores only (4) at $t = 0$.

4.2 Periodic Tanaka-profile sums

The three main Tanaka structural groups contain $m = 1, 2, 3$ component profiles. They use

$$U_h \sim \text{Unif}[0, 1], \quad h = (0.01)^{1-U_h} (0.30)^{U_h}, \quad A \sim \text{Unif}[0.10, 0.35],$$

where $A = \sum_{j=1}^m \alpha_j$ and α_j is the amplitude-to-depth ratio of component j . For $m > 1$, A is divided by a uniform simplex split. The steep structural group has one component and uses

$$U_h \sim \text{Unif}[0, 1], \quad h = (0.20)^{1-U_h} (0.35)^{U_h}, \quad \alpha_1 \sim \text{Unif}[0.25, 0.45].$$

Let

$$q = \#\{j : s_j = +1\}$$

be the number of right-moving crests. For a main case, each pair (m, q) with $m \in \{1, 2, 3\}$ and $q = 0, \dots, m$ is a parameter category. These give $2 + 3 + 4 = 9$ categories. The steep one-crest group contributes the two categories $q = 0, 1$, for eleven Tanaka categories in total. Conditional on q , the sampler chooses uniformly which q of the m exchangeable crest labels move right; the rest move left.

The accepted-case quota is divided as evenly as possible across all eleven categories. Therefore the distinct direction compositions receive equal coverage, rather than the binomial frequencies generated by independent fair signs. Left and right directions remain balanced marginally in the ideal equal-category population. For a finite quota, category counts differ by at most one and a one-case direction imbalance is possible; the fixed order gives a

quota of 1024 one extra left-moving one-crest case. Because the one-, two-, and three-crest main groups contain two, three, and four direction compositions, their aggregate shares are 2/11, 3/11, and 4/11; the steep group has share 2/11. This deliberately replaces the old four-group independent-sign mixture. No replacement-corpus cases have yet been generated under this new law.

Changing the category law advances the global generator revision. The direction-aware Tanaka sampler and its stored parameter record use revision 2. Revision-1 parameter smokes, dry runs, and all prior family pilot artifacts remain historical evidence for the constructor and transaction code, but they do not satisfy the revision-2 run specification and cannot be resumed into a revision-2 run. The 104-category gate and final corpus use revision 2.

These formulas interpolate geometrically between the endpoint depths. At fixed amplitude ratio, multiplying h by a constant multiplies both the height and width of a Tanaka component by that constant. Equal increments of U_h therefore represent equal multiplicative changes of profile scale. This is a numerical coverage rule, not a probability model for physical water depths.

The centers are periodic and separated. Draw nonnegative simplex weights w_j with $\sum_j w_j = 1$ and set the cyclic gaps to

$$d_j = 3h + (2\pi - 3mh)w_j. \quad (6)$$

The gaps sum to 2π and are all at least $3h$. A common uniform rotation on the periodic interval then fixes the centers x_j without favoring $x = 0$.

For each requested amplitude ratio α , the code numerically solves a unit-depth permanent-form solitary wave using Tanaka's formulation [4]. Xu and Guyenne used the same solitary-wave method for their long-time finite-depth test [3]. Let Φ be the surface-potential coordinate used in the solve, X the dimensionless horizontal coordinate, $Q = e^\tau$ the fluid-speed magnitude, τ its logarithm, θ the angle of the free-surface tangent, and F_α the resulting Froude number. The code adjusts the crest value of Q by bisection until the reconstructed crest height is α , iterates Tanaka's relations, and reconstructs the dimensionless elevation $\bar{\eta}_\alpha$ from

$$\frac{dX}{d\Phi} = \frac{\cos \theta}{Q}, \quad \frac{d\bar{\eta}_\alpha}{d\Phi} = \frac{\sin \theta}{Q}.$$

It uses 257 uniformly spaced nodes on $0 \leq \gamma \leq 2.5$, reflects them about zero to obtain 513 nodes, and applies the project-fixed map $\Phi = 0.01\gamma + \gamma^5$. It performs 80 fixed-point updates. These discretization choices are ours; they are not parameters prescribed by Tanaka.

The resulting values and tangents are joined by cubic Hermite interpolation, then extended by zero with continuous value and first derivative. Every periodic copy that intersects the domain is included. This compact extension, Hermite placement, periodization, multi-crest superposition, and spectral projection are project construction choices. In particular, a sum of profiles is an initial condition made from Tanaka solitary waves, not an exact steady multi-soliton solution. For component j ,

$$\eta_j(x) = h \sum_{r \in \mathbb{Z}} \bar{\eta}_{\alpha_j} \left(\frac{x - x_j + 2\pi r}{h} \right). \quad (7)$$

Only finitely many terms are nonzero. The interpolation is evaluated first on an eight-times finer grid and then restricted to the 1024-point grid.

Set $c_j = F_{\alpha_j} \sqrt{gh}$, with $g = 1$ in this corpus, and define

$$R_j(x) = (1 + \eta_{j,x}(x)^2)(c_j^2 - 2\eta_j(x)). \quad (8)$$

The following conversion of the traveling profile into the project's periodic Zakharov variables is not quoted as an equation from Tanaka. The candidate derivative of the surface potential is

$$r_j(x) = s_j [c_j - \sqrt{R_j(x)}].$$

A periodic antiderivative exists after subtracting the spatial mean of r_j . The program therefore defines ξ_j by

$$\partial_x \xi_j = r_j - \frac{1}{2\pi} \int_0^{2\pi} r_j(x) \, dx, \quad \int_0^{2\pi} \xi_j(x) \, dx = 0,$$

using Fourier division by ik on every nonzero mode. Finally,

$$(\eta_0, \xi_0) = P_{128} \sum_{j=1}^m (\eta_j, \xi_j).$$

A finite negative value of R_j is outside this construction and yields a zero-row attempt; it is never clipped to zero. A nonfinite value or an invalid wave speed signals a construction error and stops generation rather than being silently redrawn.

4.3 Benjamin–Feir carrier and sidebands

This family uses depth $h = 5$. Let ϵ_c denote carrier steepness. Choose a carrier mode $n_c \in \{4, \dots, 20\}$ and a positive sideband separation $\Delta n < n_c$. The leading instability condition is

$$0 < \frac{\Delta n/n_c}{2\sqrt{2}\epsilon_c} < 1, \quad (9)$$

Exactly 66 integer pairs $(n_c, \Delta n)$ can satisfy (9) with $0.05 \leq \epsilon_c \leq 0.13$; each pair is one parameter category. The condition is the leading Benjamin–Feir sideband band, written in wavenumber variables [5]. Conditional on the pair, draw

$$\epsilon_c \sim \text{Unif}(\epsilon_{\min}, 0.13], \quad \epsilon_{\min} = \max\left\{0.05, \frac{\Delta n}{2\sqrt{2}n_c}\right\},$$

together with a sideband ratio $\rho \sim \text{Unif}[0.05, 0.20)$ and a common sideband phase $\phi \sim \text{Unif}[0, 2\pi)$.

Because the period is 2π , set $k_c = n_c$, $k_{\pm} = n_c \pm \Delta n$, and $a = \epsilon_c/k_c$. Let (η_c, ξ_c) be the fixed and recorded fifth-order deep-water Stokes carrier. Its phase is fixed to zero as a translation convention. The initial state is

$$\begin{aligned} \eta_0(x) &= \eta_c(x) + \rho a \{\cos(k_-x + \phi) + \cos(k_+x + \phi)\}, \\ \xi_0(x) &= \xi_c(x) + \rho a \left\{ \sqrt{\frac{1}{k_-}} e^{k_- \eta_0(x)} \sin(k_-x + \phi) + \sqrt{\frac{1}{k_+}} e^{k_+ \eta_0(x)} \sin(k_+x + \phi) \right\}, \end{aligned} \quad (10)$$

followed by removal of the mean of ξ_0 and projection to the delivered band. The sidebands follow equation (33) of Xu and Guyenne, who in turn followed Dommermuth and Yue [3, 6]. Their single published case used

$$(n_c, \Delta n, \epsilon_c, \rho, \phi) = (9, 2, 0.13, 0.10, -\pi/4)$$

and a numerically computed steady deep-water carrier. Our code preserves their symmetric Airy-sideband formula, common phase, and use of the total η_0 inside both exponentials. The 66 mode-pair categories, randomized parameters, finite proxy depth $h = 5$, and analytic fifth-order carrier are our generalization, so this is not an exact reproduction of their one initial state.

4.4 Windowed JONSWAP/TMA random seas

Each random-sea parameter category fixes one depth regime, one peak-enhancement value $\gamma \in \{1, 3.3, 5\}$, and one right-moving energy fraction $r_d \in \{0, \frac{1}{2}, 1\}$. This gives $3 \cdot 3 \cdot 3 = 27$ categories.

Let H_s denote significant wave height. In the shallow regime, choose a reference peak mode $n_p \in \{16, 18, 20, 22, 24\}$. Define the relative peak depth $\chi_p = k_p h$ and the relative significant height $\delta_s = H_s/(2h)$, then draw

$$\chi_p \sim \text{Unif}[0.2, 1.5], \quad \delta_s \sim \text{Unif}[0.03, 0.16]$$

uniformly over the part of the rectangle satisfying $\chi_p \delta_s \leq 0.15$. Set $k_p = n_p$, $h = \chi_p/k_p$, and $H_s = 2h\delta_s$. In the finite-depth regime, draw

$$k_p \in [2, 12], \quad h \in [0.1, 1.5], \quad H_s \in [0.005, 0.03]$$

independently and uniformly. The deep regime uses the same k_p and H_s ranges with $h \in [5, 25]$.

The remaining construction is deterministic once independent right- and left-moving phase arrays have been drawn uniformly on $[0, 2\pi)$. For wavenumber $k > 0$, define

$$\omega(k, h) = \sqrt{k \tanh(kh)}$$

and let $\omega_p = \omega(k_p, h)$. The input k_p is a reference peak parameter; after finite-depth modification, cell integration, and tapering, the largest discrete cell need not be centered exactly at k_p . Define $\sigma = 0.07$ for $\omega \leq \omega_p$ and $\sigma = 0.09$ for $\omega > \omega_p$. The unnormalized JONSWAP frequency density is [7]

$$\tilde{S}_J(\omega) = g^2 \omega^{-5} \exp\left[-\frac{5}{4} \left(\frac{\omega_p}{\omega}\right)^4\right] \gamma^{\exp[-(\omega - \omega_p)^2 / (2\sigma^2 \omega_p^2)]}. \quad (11)$$

The factor $g^2 = 1$ could be omitted numerically because the density is normalized below, but it is retained here to display the published spectral shape. The TMA finite-depth correction of Bouws et al. [8] is

$$\Phi(kh) = \frac{\tanh^2(kh)}{1 + 2kh/\sinh(2kh)}. \quad (12)$$

Thus the frequency density used is $\tilde{S}_J(\omega)\Phi(kh)$. This reduces low-frequency energy in shallow water and tends to the ordinary JONSWAP form in deep water.

For each positive integer mode $j \leq 128$, define the group velocity $c_g(k, h) = \partial\omega(k, h)/\partial k$. The program uses fixed 16-point Gauss–Legendre quadrature and the window

$$W(k) = \begin{cases} 1, & 0 \leq k \leq 96, \\ \cos^2\left[\frac{\pi}{2} \frac{k - 96}{32}\right], & 96 < k < 128, \\ 0, & k \geq 128. \end{cases} \quad (13)$$

to compute the nonnegative cell energy

$$E_j = \int_{j-1/2}^{j+1/2} \tilde{S}_J(\omega(k, h)) \Phi(kh) c_g(k, h) W(k) dk.$$

This is the frequency density converted to a wavenumber density and integrated over one Fourier cell. Normalize the cell energies by $w_j = E_j / \sum_{\ell=1}^{128} E_\ell$. The prescribed surface-elevation variance is $m_0 = (H_s/4)^2$. Set

$$A_{j,+} = \sqrt{2r_d m_0 w_j}, \quad A_{j,-} = \sqrt{2(1 - r_d) m_0 w_j}.$$

For the stored phases $\phi_{j,+}$ and $\phi_{j,-}$, the initial state is

$$\begin{aligned} \eta_0(x) &= \sum_{j=1}^{128} \{A_{j,+} \cos(jx + \phi_{j,+}) + A_{j,-} \cos(jx + \phi_{j,-})\}, \\ \xi_0(x) &= \sum_{j=1}^{128} \frac{\omega(j, h)}{j \tanh(jh)} \{A_{j,+} \sin(jx + \phi_{j,+}) - A_{j,-} \sin(jx + \phi_{j,-})\}. \end{aligned} \quad (14)$$

No realization-dependent rescaling or visual roughness test is applied. The parameter strata, $\gamma \in \{1, 3.3, 5\}$, direction split, independent phases, Fourier-cell quadrature, and 96–128 window are project choices. The cited papers provide the continuous JONSWAP and TMA spectral factors, not these corpus-balancing and discretization rules.

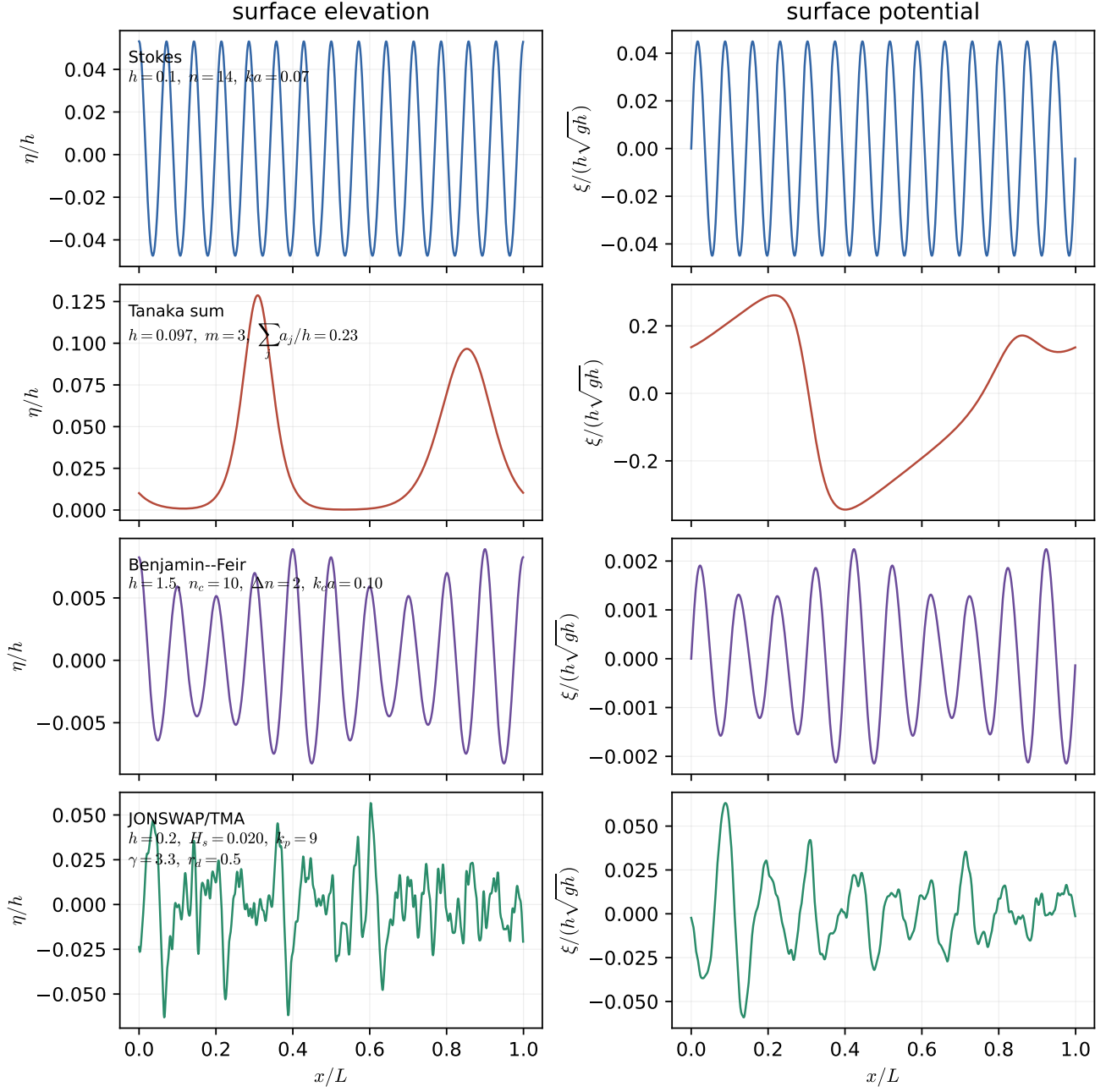


Figure 1: One constructed initial condition from each family. The left column shows η/h , and the right column shows $\xi/(h\sqrt{gh})$, with $g = 1$. These are examples of the formulas above, not cases selected by model error.

5 One reference rollout and one whole-trajectory decision

Stokes cases are static. Tanaka, Benjamin–Feir, and random-sea cases are evolved with the same order-six DNO. Define the internal output

$$G_{\text{int}}(\eta; h)\xi = \mathcal{G}_6^{1024,8}(P_{128}\eta; h)(P_{128}\xi).$$

The discrete Zakharov equations used for the rollout are

$$\begin{aligned}\eta_t &= G_{\text{ref}}(\eta; h)\xi, \\ \xi_t &= P_{128}^\circ \left[-\eta - \frac{1}{2}\xi_x^2 + \frac{(G_{\text{int}}(\eta; h)\xi + \eta_x \xi_x)^2}{2(1 + \eta_x^2)} \right].\end{aligned}\tag{15}$$

The Zakharov surface formulation and the Craig–Sulem expansion follow the formulation used by Xu and Guyenne [3]. The truncation order, spatial grid, padding factor, and delivered Fourier band in this note are fixed project choices. The Craig–Sulem calculation inside G_{int} uses padding factor eight. The products written explicitly in the second line of (15) are formed on a separate two-times Fourier grid before projection.

The linear gravity-wave part of (15) is advanced exactly in Fourier space. More precisely, write the right-hand side as $L_h z + R_h(z)$, where $z = (\eta, \xi)$, L_h is the linear part, and R_h is the remainder. The change of variable $v(t) = e^{-tL_h} z(t)$ gives

$$v_t = e^{-tL_h} R_h(e^{tL_h} v).$$

The program applies the two-stage, fourth-order Gauss–Legendre method (GL2) to this transformed equation. This is the integrating-factor GL2 method described by Xu and Guyenne [3]. The step, iteration cap, and residual test below are our fixed implementation settings.

The GL2 step is 0.01. At each step it solves two coupled internal stage equations by at most four fixed-point updates. For each stage and each field separately, the remaining stage-equation mismatch is divided by the larger of the stage norm, the fixed-point-map norm, and the smallest normal floating-point number. The largest of these relative mismatches must be at most 10^{-8} . After a completed step the two state fields are projected back to their prescribed Fourier bands. A candidate state is stored every 0.08, exactly every eight steps; there is no time interpolation. The step 0.01 was selected before production on a fixed 12-case panel. All ten cases that completed at both steps 0.01 and 0.005 agreed to at most 1.126337×10^{-5} ; the other two became incomplete under both steps. The 0.005 calculation is therefore a method-level check, not a second rollout applied to each production case.

Tanaka and Benjamin–Feir request $0 \leq t \leq 200$. A random sea uses the peak period

$$T_p = \frac{2\pi}{\omega_p}$$

and requests the last 0.08-grid time no later than $16T_p$. Cases with different random-sea horizons may share a numerical batch, but each one is judged only on its own requested prefix.

Suppose the initial condition already obeys its family’s parameter rules. There is then one trajectory decision:

$$\boxed{\text{accept} \iff \text{the fixed method returns one complete admissible trajectory.}}\tag{16}$$

Here *complete admissible trajectory* means that the calculation reaches its requested final time, every GL2 stage has finite relative residual at most 10^{-8} , every stage and updated state is finite, all saved η , ξ , and $G_{\text{ref}}(\eta; h)\xi$ values are finite, and

$$h + \eta(x_\ell, t_j) > 0$$

at every saved time t_j and grid point x_ℓ . The final inequality is the elementary geometric requirement that the graph of the free surface remain above the flat bottom $y = -h$. It has already been checked at $t = 0$ while assembling the initial batch and is checked again on every saved state. The recorded diagnostics state which clause failed; they are not additional fitted filters. A failed trajectory contributes zero rows. Keeping its finite prefix would systematically give difficult cases only their earlier, easier states.

Only the one $\Delta t = 0.01$ trajectory is run for each production case. Sign changes, slope, extrema, spectrum, Hamiltonian drift, absolute target size, visual appearance, and model error do not decide acceptance.

A static Stokes state has no trajectory. It is kept only when its sampled parameters obey the Stokes rules, its state and common target are finite, and $h + \eta_0 > 0$ at every grid point.

5.1 Endpoint diagnostics that do not reject cases

Spatial roughness and sign changes remain useful for finding cases that merit inspection, even though neither quantity decides acceptance. We applied two such rankings to the fixed method-level rollout panel. We first removed the two trajectories that did not reach their requested terminal time, exactly as the corpus rule requires. For the figures below, we then omit the three JONSWAP/TMA cases because their intentionally broadband spectra otherwise dominate both rankings and obscure comparison among the coherent-wave families. This second omission is only a presentation choice; it does not remove any case from the corpus. The displayed rankings therefore use five completed Tanaka and Benjamin–Feir endpoints. No last-finite frame from an incomplete trajectory participates.

Let $\hat{\eta}_k$ be the Fourier coefficients of an endpoint elevation after removal of its mean. Define its root-mean-square wavenumber by

$$k_{\text{eff}}(\eta) = \left(\frac{\sum_{k \neq 0} k^2 |\hat{\eta}_k|^2}{\sum_{k \neq 0} |\hat{\eta}_k|^2} \right)^{1/2}. \quad (17)$$

This is large when a substantial fraction of the elevation energy lies at short spatial scales; by convention it is zero when the denominator vanishes. Figure 2 shows all five completed non-JONSWAP/TMA cases in decreasing order.

For the second ranking, let $f_\ell = [G_{\text{ref}}(\eta; h)\xi](x_\ell)$ at the endpoint and form the cyclic forward differences

$$d_\ell = f_{\ell+1} - f_\ell, \quad \ell = 0, \dots, 1023, \quad f_{1024} = f_0.$$

Set $\tau = 0.03 \max_\ell |d_\ell|$, discard differences with $|d_\ell| \leq \tau$, and count cyclic transitions between the signs of the remaining differences. Denote this count by $C_{0.03}(G_{\text{ref}}(\eta; h)\xi)$. The dead zone prevents roundoff-sized differences from creating sign changes. Figure 3 shows all five completed non-JONSWAP/TMA cases.

Completed non-JONSWAP endpoints: diagnostic rankings, not rejection rules

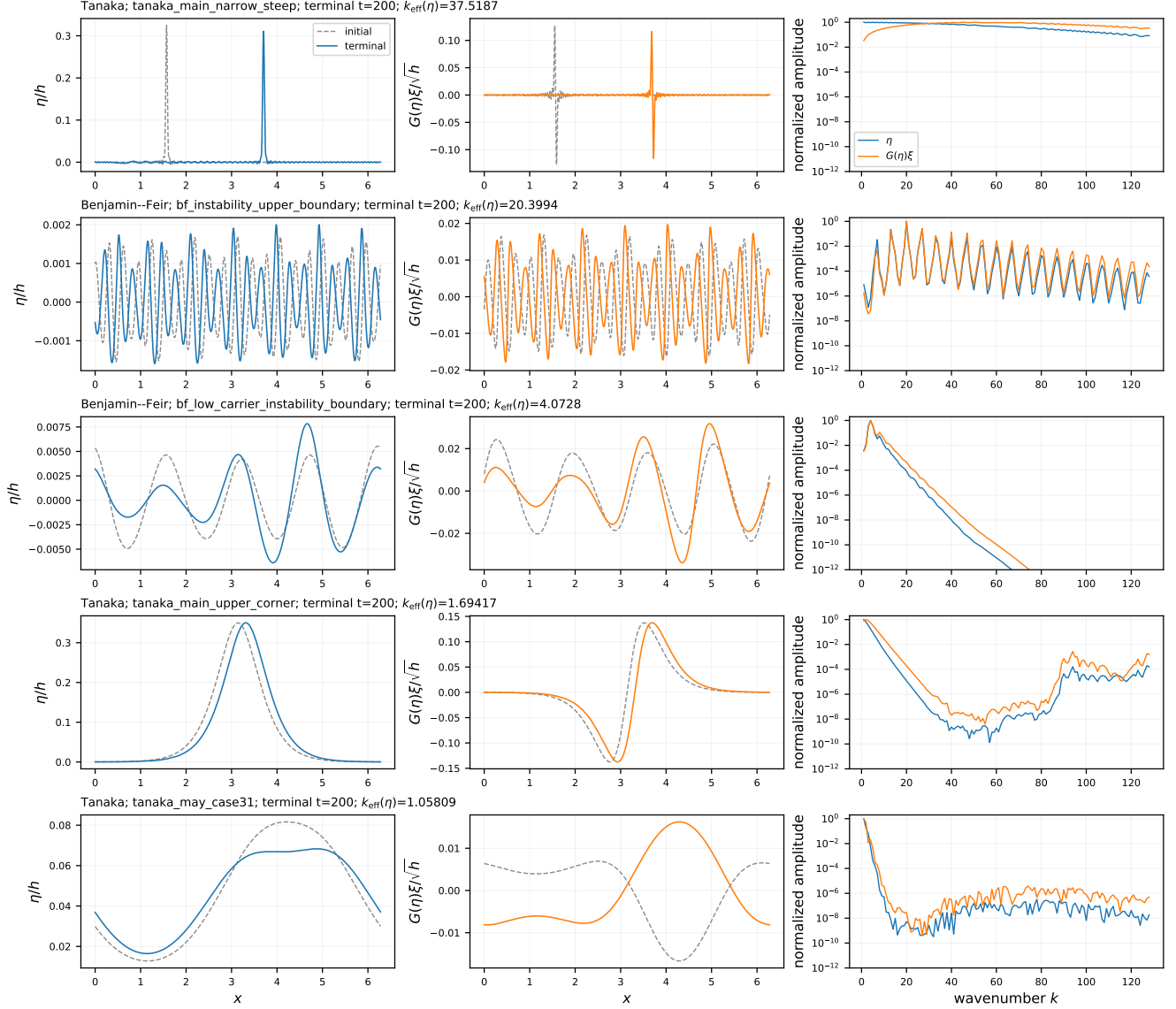


Figure 2: Cases with the largest endpoint value of $k_{\text{eff}}(\eta)$. In each row, the dashed gray curve is the initial field, the colored curve is the terminal field, and the right panel shows the corresponding normalized Fourier amplitudes. Every displayed trajectory completed its requested interval, and JONSWAP/TMA cases are omitted from this display. The narrow Tanaka case has the largest value, $k_{\text{eff}} = 37.52$; spatial roughness alone is therefore not evidence of numerical failure.

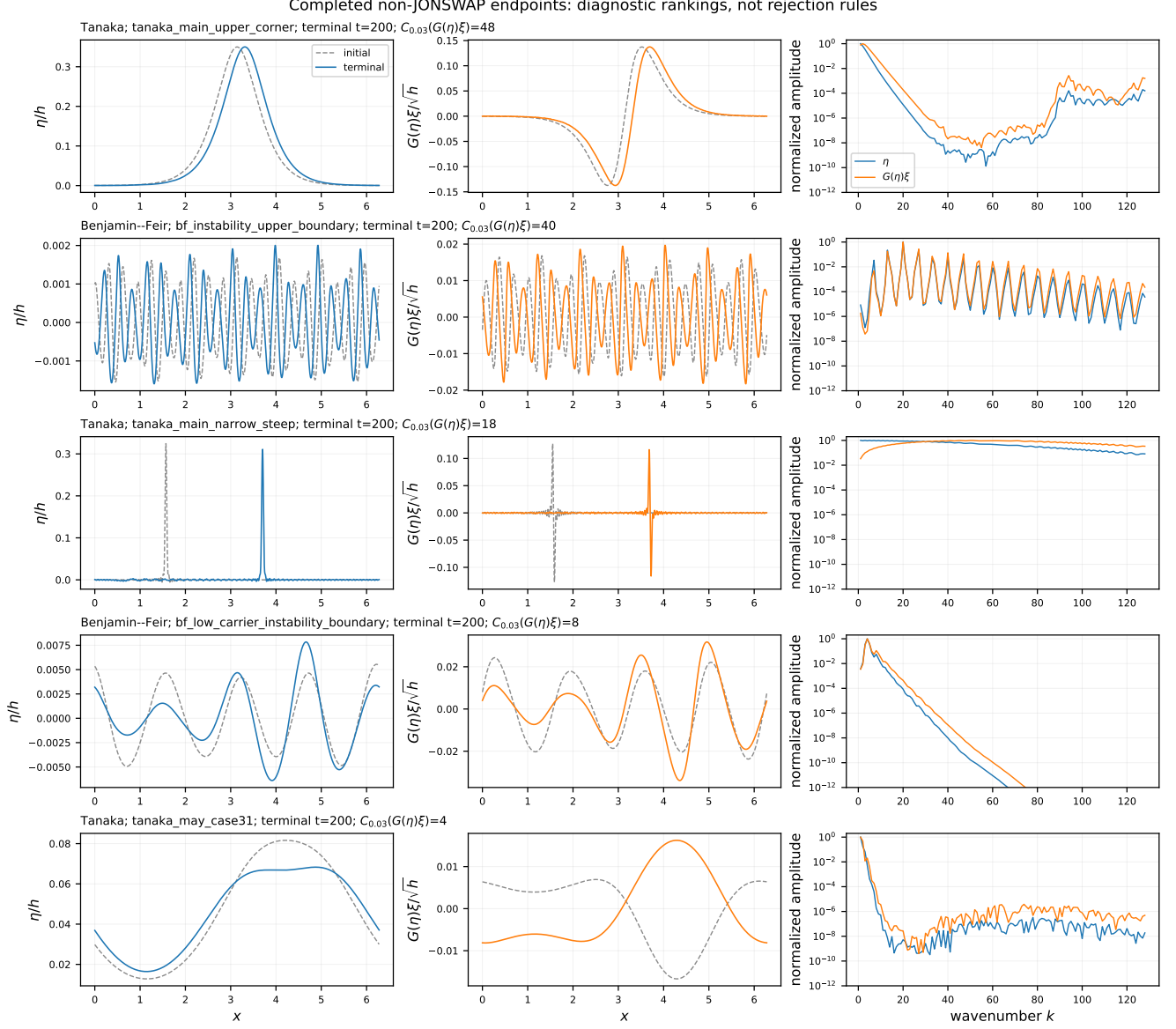


Figure 3: Cases with the largest endpoint sign-change count in the cyclic forward difference of $G_{\text{ref}}(\eta; h)\xi$, after incomplete trajectories and the broadband JONSWAP/TMA display cases have been removed. The completed Tanaka upper-corner and Benjamin-Feir upper-boundary cases lead with 48 and 40 changes, followed by the narrow Tanaka case with 18. The count detects small oscillatory contributions but does not by itself show that a completed trajectory is numerically unstable, so it is retained only as a diagnostic.

6 Selecting times after acceptance

Time selection reduces storage; it never decides whether a case is accepted and it never uses model error. Stokes retains only $t = 0$. Tanaka and Benjamin–Feir have 2501 candidate times

$$t_j = 0.08j, \quad j = 0, \dots, 2500,$$

and retain exactly 200 distinct times, including both endpoints.

For Tanaka, define the total surface-slope energy

$$S_{T,j} = \sum_{\ell=0}^{1023} |\eta_x(x_\ell, t_j)|^2$$

and its discrete time difference

$$DS_{T,j} = \begin{cases} S_{T,1} - S_{T,0}, & j = 0, \\ (S_{T,j+1} - S_{T,j-1})/2, & 1 \leq j \leq 2499, \\ S_{T,2500} - S_{T,2499}, & j = 2500. \end{cases}$$

The Tanaka activity is

$$A_{T,j} = \frac{|DS_{T,j}|}{S_{T,j} + 10^{-12}}.$$

Thus extra weight is given to times when the surface shape changes rapidly.

For Benjamin–Feir, define

$$S_{B,j} = \max_{0 \leq \ell < 1024} |\eta(x_\ell, t_j)|, \quad A_{B,j} = \left(S_{B,j} - \min_{0 \leq r \leq 2500} S_{B,r} \right)^4.$$

This emphasizes large surface-elevation excursions; it is the maximum absolute elevation, not an analytic-signal envelope.

Let $F = T$ denote Tanaka and $F = B$ denote Benjamin–Feir. Each activity sequence $A_{F,j}$ is smoothed by a fixed Gaussian in the time index. Its standard deviation is 50 candidate times for Tanaka and 20 for Benjamin–Feir. The kernel stops at three standard deviations, and values beyond either endpoint are replaced by that endpoint value. If $\tilde{A}_{F,j}$ is the smoothed activity, the selection weights are

$$p_{F,j} = \frac{1 - \alpha_F}{2501} + \alpha_F \frac{\tilde{A}_{F,j}}{\sum_{r=0}^{2500} \tilde{A}_{F,r}}, \quad \alpha_F = \begin{cases} 0.50, & \text{Tanaka,} \\ 0.85, & \text{Benjamin–Feir.} \end{cases}$$

If all activities vanish, the second term is replaced by a uniform weight. The program takes fixed midpoint quantiles of the cumulative weights, pins the first and last times, and clips each quantile index so that enough unused indices remain for all later quantiles. A forward pass then advances any repeated index to the next unused one. The result is deterministic.

For a random sea, let

$$T_s = 0.08 \left\lfloor \frac{16T_p}{0.08} \right\rfloor, \quad J_s = T_s/0.08.$$

It retains the 16 indices

$$j_\ell = \left\lfloor \frac{\ell J_s}{15} + \frac{1}{2} \right\rfloor, \quad \ell = 0, \dots, 15.$$

These are approximately uniform and include 0 and T_s .

7 Identity, splitting, quotas, and storage

7.1 Case identity and data splits

The full case identity consists of

(family, revision, split, stream, attempt).

Its deterministic random stream instead uses these five integer seed words:

(split root, family, revision, stream, attempt).

These words seed NumPy’s PCG64 generator. The split roots are

split	training	validation	test
root	2026072210	2026072204	2026072205.

The split, parameter category, and entire sampled specification are written before construction or integration. A rejection remains attached to that split. All stored times from one trajectory therefore stay together, so nearby times from the same numerical solution cannot cross into validation or test.

7.2 Accepted-case quotas

A quota is a required number of accepted complete cases, not a number of attempts or stored rows. Suppose a family has m ordered parameter categories and requires C accepted cases. Write

$$C = bm + r, \quad 0 \leq r < m.$$

The first r categories receive $b + 1$ cases and the other categories receive b . This is the most even integer division possible.

A failed attempt is replaced only in its original category. A category assigned Q accepted cases may use at most $4Q$ recorded attempts. If the resolved attempts reach this bound without filling Q , the family run stops as incomplete. Another category cannot fill the deficit. The factor four only bounds computation; it does not change any case-level decision. The Stokes amplitude redraws described above are internal to one attempt.

7.3 Files and combined view

An accepted row stores three length-1024 spatial arrays η , ξ , and $G_{\text{ref}}(\eta; h)\xi$ in float32, together with float64 depth and time, its selected candidate-time index, and its case-local index in that array file. The case record preserves the sampled parameters and numerical decision for both accepted and rejected attempts. It also binds the family, split, random stream, numerical settings, software versions, source hashes, and array checksum.

After the four families have been generated for all three splits, the combined builder checks the records, shapes, checksums, numerical settings, and required counts. It then writes a compact row-to-case index pointing to the existing arrays rather than copying them. This index is the dataset view read by the trainer.

8 Production size and trainer handoff

The first checkpoint requests, from *each* family,

$$2048 \text{ training,} \quad 1024 \text{ validation,} \quad 1024 \text{ test cases.}$$

The four families therefore contain 16,384 accepted cases in total. For one accepted case from each family, the stored-row count is

$$1 + 200 + 200 + 16 = 417.$$

The exact split counts are The current uncompressed estimate for the field archives and row map is 19.62 GiB.

Split	Accepted cases	Stored rows
Training	8,192	854,016
Validation	4,096	427,008
Test	4,096	427,008
Total	16,384	1,708,032

Training then grows by additive, disjoint chunks to

$$C_{\text{tr}} \in \{2048, 4096, 8192, 16384\}$$

accepted cases per family; validation and test remain fixed at 1024 cases per family. The resulting cumulative sizes are

Training cases per family	Total stored rows
2,048	1,708,032
4,096	2,562,048
8,192	4,270,080
16,384	7,686,144

Thus 2048 is a checkpoint, not the completed corpus. The target at 16,384 contains 73,728 accepted cases across the four families and three splits. Its 7,686,144 rows are close to the 7,633,989 rows in historical v9: 52,155 more rows, or 0.68%.

Normalization is computed from all stored *training* rows and from no validation, test, or diagnostic rows. Let η_i , ξ_i , and h_i denote the elevation, potential, and depth in training row i , and let $G_i^* = G_{\text{ref}}(\eta_i; h_i)\xi_i$ denote its reference label. Define

$$s_\eta = \max_{i,x} |\eta_i(x)|, \quad s_\xi = \max_{i,x} |\xi_i(x)|, \quad s_G = \max_{i,x} |G_i^*(x)|.$$

The intended corpus gives positive values for all three scales, and the release audit verifies this before training. The current loader uses one as a defensive fallback if a nonpositive scale reaches it; that fallback is not treated as validation of a malformed corpus. The cached values are tied to the checked dataset identity. The trainer receives

$$\eta/s_\eta, \quad \xi/s_\xi, \quad \log(h/h_\star), \quad G_{\text{ref}}(\eta; h)\xi/s_G,$$

where the reference length is $h_\star = L/(2\pi) = 1$ on the production domain. The paper-corpus training command must explicitly select this scale normalization with `-norm scale`; the trainer’s generic min–max default is not the rule stated here.

At every training epoch, the loader chooses one stored time uniformly from every accepted training case. Thus each family contributes C_{tr} samples per epoch even though the archive stores different numbers of times per case. Validation uses one fixed deterministic time per case. Airy, manufactured-DNO, and sparse-Fourier states remain diagnostic tests and do not enter training or its normalization.

9 Checks before bulk generation

Grid, truncation, padding, spectral window, and time step are chosen on fixed method-level panels before bulk generation. They are not re-estimated for individual production cases. The final execution check asks for one accepted case, using all production settings and the present single-trajectory executor, from every trajectory parameter category:

$$11 \text{ Tanaka} + 66 \text{ Benjamin-Feir} + 27 \text{ JONSWAP/TMA} = 104 \text{ categories.}$$

This *104-category pilot* uses the production solver, trial batch size four, and at most four attempts per category. It is kept separate from the final corpus. The four static Stokes categories have already passed their exact calculation under generator revision 2: all four cases were accepted and all four quality-decision masks were zero. The pilot asks whether every declared trajectory category can be constructed and evolved; one case per category does not estimate a population rejection rate.

Bulk generation begins only after this pilot completes, its wall time and peak memory are recorded, and its result determines the production batch size. The final audit checks exact case and row counts, disjoint splits, finite stored arrays, positive depths, zero-row failed cases, checksums, and training-only normalization.

Present status. As of 27 July 2026, exact generator-revision-2 calculations have passed for the four static Stokes categories, and the 104-category trajectory pilot is running. Bulk generation remains conditional on that pilot passing. Therefore the current single-trajectory contract is not yet validated across all trajectory parameter categories; the 7,686,144-row target corpus has not been generated; its population acceptance rates and runtime are unknown; and no reported model has been trained on it. This note is the frozen construction contract, not a report of a completed replacement dataset.

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